

July 10, 2025

Electrons, Empires, and the Elemental Race for Copper

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I had been working on my next *AI Macro Nexus* write-up about copper on Tuesday when the tariff news broke. Not surprisingly, since the announcement, there have been many inquiries about ways to play copper. Colin Fenton has been highlighting the deepening bullish setup in copper since August 2024. He reiterated this view at the end of 2024, when investor sentiment was at decade lows—even as physical market conditions were tightening rapidly. That contrarian call has since been validated by a 17% year-to-date rally in the Global X Copper Miners ETF and double-digit gains across major futures markets.

But even those price moves may understate the deeper shift underway. I'll leave the near term demand and supply numbers to Colin, but I want to focus on the importance of copper over the next decade. This copper bull market isn't about a cyclical PMI recovery or a China reacceleration; it's about the structural reality of the AI buildout and the electrification needed to support it. This is a geopolitical race for military supremacy and a hyperscaler race to avoid obsolescence.

In a world conditioned based on the industrial revolution to expect boom-bust cycles and speculative bubbles, it's easy to mistake structural shifts for temporary spikes but this is no AI bubble, it's a certainty. The geopolitical race for military supremacy and the Magnificent Seven's fear of obsolescence are driving an AI infrastructure arms race that makes this transformation in copper demand undeniably structural. In this note, I want to expand on that side of the story: how AI, robotics, and grid-scale compute are fundamentally transforming the copper demand curve and doing so with a level of certainty that few macro trends in history can match.

A Once-in-a-Century Setup

In May 2025, Elon Musk's xAI began scaling its "Colossus" data center campus in Memphis, a facility projected to house over 200,000 GPUs and draw more than 150 megawatts from a new dedicated substation. The energy intensity of these AI "compute factories" has become emblematic of a broader reality: the AI boom is not just a digital phenomenon, but a physical one. Beneath every model, inference, and humanoid deployment lies an unprecedented demand for electricity and with it, copper.

What makes this moment so distinct is the certainty of the buildout. Unlike most historical infrastructure or technology waves, the AI race is a convergence of geopolitical urgency, economic inevitability, and military modernization. It combines the Manhattan Project's existential pressure, the Space Race's national prestige, the Industrial Revolution's transformational scope, and the Internet boom's commercial energy. Yet it exceeds them all in scale and speed. Governments are mandating AI deployment across civilian and defense sectors, hyperscalers are investing hundreds of billions in chips and data centers, and military institutions are racing to integrate humanoids and autonomous systems into force structure. This is not a market guess, it is a full-spectrum mobilization that touches infrastructure, labor, energy, and national security. And copper is the critical thread linking all of it.

Copper is entering the most structurally certain and materially demanding bull market in modern commodity history. Unlike past cycles that depended on boom-and-bust debt driven dynamics or fragile geopolitical narratives, the next wave of copper demand is being driven by deeply embedded global transitions: electrification and digitalization. What makes this moment unique is not just the scale of demand, but the clarity with which we can see it coming. From AI data centers and electric vehicles to national grid modernization and robotics, copper is at the heart of the infrastructure required to power the next chapter of human and economic progress. It is not speculative, but infrastructural. It is not cyclical, but structural.

The Certainty of Electrification-Driven Demand

The surge in copper demand is not a theoretical projection—it is a policy-locked certainty. AI data centers are projected to consume up to 25% of U.S. electricity by 2030, a fivefold increase from today, requiring tens of thousands of tonnes of copper per hyperscale facility. Importantly, AI and high-performance computing data centers are materially more copper-intensive per megawatt than conventional compute infrastructure. A traditional enterprise data center might use 30–50 tonnes of copper per 10 MW of capacity. In contrast, next-generation AI “compute factories” can require 80–100 tonnes per MW, driven by denser power distribution, liquid cooling systems, and internal copper cabling between high-performance chips. This means that even small increases in AI workload translate into large step-changes in copper demand. Grid expansion is a global imperative: BloombergNEF forecasts that transmission and distribution networks will double in size by 2050, driving copper demand from grids alone to 13 Mt annually by 2030. Add to this the electric vehicle transition, where a single EV uses 3–5 times more copper than a combustion vehicle, and the rollout of humanoid robots, which could add hundreds of thousands of tonnes per year in incremental copper usage. This is not demand driven by consumer trends but by foundational rewiring of how economies function.

Supply Cannot Scale Fast Enough

While demand is locked in, copper supply is anything but. Global mine production was just 21.8 Mt in 2022, and many of the world’s largest copper deposits are aging, with ore grades declining by 40% since 1990. New discoveries are rare and permitting timelines average 15–20 years. Political instability in key producing countries like Chile and Peru, water constraints in mining regions, and ESG pressures further restrict supply. Even in a best-case scenario, the world faces a 6–10 Mt annual supply gap by the 2030s equivalent to three new “Escondidas” per year, which is logistically and geologically implausible. This imbalance sets the stage for persistent structural deficits.

Why This Time Is Different: Peak Copper vs. Peak Oil

In 2007, fears of “peak oil” captured the public imagination. But that crisis was solved through the GFC economic disruption, the U.S. shale breakthroughs and China’s fix asset investment slowdown. Copper is fundamentally different. Oil has substitutes and elasticity on both supply and demand sides; copper does not. There is no “shale copper,” and there currently are no alternative for conducting electrons at scale. I say currently because I am sure AI will help with all of this eventually but not based on today’s information and situation. More importantly, global electricity demand is set to re-accelerate after 50 years of stagnation in developed economies. AI, EVs, robotics, and heat pumps are driving the fastest and most globally distributed rise in electricity use since the post-war period. Grid buildouts, AI server loads, and electrified transport all converge on one material: copper. This is not a repeat of peak oil, this is the emergence of a systemic constraint.

The Digitalization Multiplier

Digital technologies are not only creating more demand for electricity they are making that demand less deferrable and more copper intensive. Each AI inference, each robot in a factory, each sensor in a smart building adds marginal copper usage in power delivery, cooling, and connectivity. The growth in token demand from AI is effectively a proxy for electricity usage growth, which is a proxy for copper. Unlike past demand waves, which were tied to population or GDP growth, this wave is tied to digital scale which can expand tenfold without any corresponding rise in population. As AI becomes ubiquitous, copper becomes the physical infrastructure beneath the intelligence.

Strategic and Investment Implications

The implications are profound. Investors should treat copper not as a cyclical trade, but as a core structural allocation. Governments must begin treating copper as a strategic resource on par with oil in the 20th century. Policy actions such as strategic stockpiling, permitting reform, and bilateral supply agreements are not optional they are essential. Meanwhile, companies in sectors from mining to manufacturing must prepare for an era where copper is the bottleneck. Copper ETFs, royalty firms, and low-cost producers in stable jurisdictions stand to benefit disproportionately. There is no version of the electrified, digitized future that does not run through copper.

The Red Metal’s Golden Era

Copper is no longer just an industrial input, it is the foundational material of the 21st century. Its demand profile is clear, its supply profile constrained, and its relevance growing. This is a rare moment in the history of commodities: a super-cycle that is both globally diversified and structurally locked in. For investors, governments, and innovators alike, understanding and acting on the copper bottleneck will be one of the defining strategic decisions of the coming decade. Investor sentiment is currently subdued despite copper's resilience because the U.S. manufacturing PMI has been below 50 in all but two months since November 2022. I believe that situation is about to change on the back of the AI buildout and if so, copper will become a broader bullish story. The red metal is entering its golden era not as a speculative trade, but as the beating heart of an electrified and intelligent global economy.

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